Robert Riley Nerem

San Diego, CA | 303-880-4420 | riley.nerem@gmail.com | robertrileynerem.com

RESEARCH

Research in neural algorithmic reasoning, including generalization on graphs, combinatorial optimization, and algorithmic alignment. Developing theory and neural models that combine learning with classical algorithms for scientific and engineering applications.

EDUCATION

Doctor of Philosophy, Data Science (in progress) — September 2022–Present
University of California San Diego, San Diego, CA
Advisor: Prof. Yusu Wang, Halıcıoğlu Data Science Institute

Master of Science, Physics — May 2022
University of Calgary, Calgary, AB
Supervisor: Prof. Barry Sanders, Department of Physics and Astronomy
Co-supervisor: Prof. Peter Høyer, Department of Computer Science

Bachelor of Science, Physics — December 2018
Montana State University, Bozeman, MT
Honors Degree, *summa cum laude*
Minors: Computer Science, Mathematics

PUBLICATIONS * Indicates a primary author in addition to the first-listed author.

Conference Papers

S. Wittig*, A. Vasileiou*, **R.R. Nerem***, T. Stoll, F. Geerts, Y. Wang, and C. Morris. Which Algorithms Can Graph Neural Networks Learn? International Conference on Machine Learning (ICML), PMLR **306**, 135232–135307, 2026. **Spotlight; Oral**.

R.R. Nerem, S. Chen*, S. Dasgupta, and Y. Wang. Graph Neural Networks Extrapolate Out-of-Distribution for Shortest Paths. Conference on Learning Theory (COLT), PMLR **336**, 5273–5331, 2026.

R.R. Nerem, Z. Luo, A. Rafiey, and Y. Wang. Differentiable Extensions with Rounding Guarantees for Combinatorial Optimization over Permutations. Advances in Neural Information Processing Systems (NeurIPS), **38**, 2025.

A.B. Kahng, **R.R. Nerem**, Y. Wang, and C.-Y. Yang. NN-Steiner: A Mixed Neural-Algorithmic Approach for the Rectilinear Steiner Minimum Tree Problem. Proceedings of AAAI, **38**(12), 13022–13030, 2024.

D.V. Reddy, **R.R. Nerem**, A.E. Lita, S.W. Nam, R.P. Mirin, and V.B. Verma. Exceeding 95% System Efficiency Within the Telecom C-Band in Superconducting Nanowire Single Photon Detectors. CLEO: QELS—Fundamental Science, FF1A.3, 2019.

Preprints

R.R. Nerem, M. Black, A. Vasileiou, C. Morris, and Y. Wang. The Expressive Power of Random-Feature MPNNs Across Graph Sizes. arXiv:2610.12809, 2026.

R.R. Nerem, P. Singh, C. Ward, and Y. Wang. Algorithmically Aligned Neural Agglomerative Tree Construction. arXiv:2610.07271, 2026.

R.R. Nerem, P. Crawford-Kahrl, B. Cummins, and T. Gedeon. A Poset Metric from the Directed Maximum Common Edge Subgraph. arXiv:1910.14638, 2019 (revised 2020).

Workshop Contributions

Y. He, **R.R. Nerem**, T. Stoll, S. Cantürk, D. Georgiev, C. Qian, S. Wittig, F. Geerts, S. Jegelka, E. Vitercik, Y. Wang, N. Karalias, and C. Morris. Neural Algorithmic Reasoning Must Explain When Neuralization Adds Value. 3rd AI for Math Workshop, ICML, 2026.

Journal Articles

R.R. Nerem and D.R. Gaur. Conditions for Advantageous Quantum Bitcoin Mining. Blockchain: Research and Applications, **4**(3), 100141, 2023.

A. Alase, **R.R. Nerem**, M. Bagherimehrab, P. Høyer, and B.C. Sanders. Tight Bound for Estimating Expectation Values from a System of Linear Equations. Physical Review Research, **4**(2), 023237, 2022.

P. Crawford-Kahrl*, **R.R. Nerem***, B. Cummins, and T. Gedeon. Genetic Networks Encode Secrets of Their Past. Journal of Theoretical Biology, **541**, 111092, 2022.

D.V. Reddy, **R.R. Nerem**, S.W. Nam, R.P. Mirin, and V.B. Verma. Superconducting Nanowire Single-Photon Detectors with 98% System Detection Efficiency at 1550 nm. Optica, **7**(12), 1649–1653, 2020.

L.M. Smith, F.C. Motta, G. Chopra, J.K. Moch, **R.R. Nerem**, B. Cummins, K.E. Roche, C.M. Kelliher, A.R. Leman, J. Harer, T. Gedeon, N.C. Waters, and S.B. Haase. An Intrinsic Oscillator Drives the Blood Stage Cycle of the Malaria Parasite Plasmodium falciparum. Science, **368**(6492), 754–759, 2020.

R.R. Nerem and D. James. Charged Particle Pickup Tube Detector Used in Experimental Applications and Classroom Demonstrations. The Physics Teacher, **58**(3), 200–203, 2020.

E. Berry, B. Cummins, **R.R. Nerem**, L.M. Smith, S.B. Haase, and T. Gedeon. Using Extremal Events to Characterize Noisy Time Series. Journal of Mathematical Biology, **80**, 1523–1557, 2020.

RESEARCH EXPERIENCE

Computation

PhD Research Fall 2022–Present
University of California San Diego, San Diego, CA
Prof. Yusu Wang, Halıcıoğlu Data Science Institute

GNN expressivity and size generalization; differentiable permutation optimization; algorithmically aligned neural models.

Quantum Researcher Fall 2021–Fall 2022
Quantum Algorithms Institute, Surrey, BC

Analysis of quantum machine learning algorithms for medical biology.

Master's Research Fall 2019–May 2022
University of Calgary, Calgary, AB
Prof. Barry Sanders & Prof. Peter Høyer, Institute for Quantum Science and Technology

Quantum expectation-value algorithms and lower bounds; analysis of quantum methods for Bitcoin mining and pathogen identification.

Research Assistant Fall 2018–Summer 2019
Montana State University, Bozeman, MT
Prof. Tomas Gedeon & Dr. Breschine Cummins, Mathematics

Graph-based metrics for gene-expression time series and algorithms for conserved structure in evolving genetic networks.

Experiment

Research Assistant Summer 2018
National Institute of Standards and Technology, Boulder, CO
Dr. Sae Woo Nam, Faint Photonics Group

Python automation for precision measurements of single-photon detector efficiency and integrated photonic devices.

Research Assistant Fall 2016–Spring 2018
Montana State University, Bozeman, MT
Prof. Rufus Cone & Dr. Charles Thiel, Physics

Scanning Fabry–Perot interferometers and spectroscopic characterization of crystals for quantum information.

Research Assistant Summer 2015, Summer 2016
University of Colorado, Boulder, CO
Prof. Zoltan Sternovsky & Dr. David James, Institute for Modeling Plasma, Atmospheres and Cosmic Dust

Charged-particle pickup detector and photomultiplier instrumentation for measuring particle charge, velocity, and acceleration.